Descriptor Similarity Does Not Predict Array Response: Chemical Warfare Agent Simulants Tested Against Sixteen Authentic Agents

Ku Kang,[†] Jeongyun Kim,[†] Miae Kim,[†] Doo-Hee Lee,[‡] Jin Yoo,[†] and Yoon Jeong Jang[*,†]

[†]*Chemical, Biological, Radiological and Nuclear (CBRN) Defense Research Institute, Ministry of National Defense, Seoul, Republic of Korea*

[‡]*Department of Civil Engineering and Environmental Sciences, Korea Military Academy, Seoul 01805, Republic of Korea*

E-mail: yjjangchem@korea.kr

Abstract

Chemical warfare agent detection is developed and qualified on simulants, and we asked what measurement establishes that a simulant reproduces the agent it stands in for. We reanalyse the colorimetric tables of three published runs of one 29-dye array, which between them measured sixteen authentic agents spanning nerve, blood, choking, vesicant and A-series compounds at four exposure concentrations against a zero-concentration baseline in two optical channels, and compare the response patterns with the molecular-descriptor distances on which simulant selection in practice rests. The descriptor calculations and every test reported here are new. The array discriminates: in all six run-channel cells, every compound agrees with itself across concentrations more closely than with the other compounds in that cell, at the exact floor each cell's

four to seven compounds allow. Descriptor distance nonetheless fails to predict response distance. Twelve exact Mantel tests and six combinations of them leave nothing below $q = 0.05$, against a design that resolves a rank correlation of 0.67 at 80% power where the largest observed is 0.57. That null holds under every analyst choice we enumerated, including the descriptor encoding, the correlation filter, the conformer seed and the panel composition, while the descriptor-space count of valid matches is moved by all four and is therefore not a measurement. Dimethyl methylphosphonate, the standard G-agent simulant and the benchmark compound for decontamination efficiency, is anti-correlated with soman and cyclosarin in the ultraviolet channel. In the three runs reanalysed here, one designated simulant has been measured beside the agents.

Introduction

Nearly every chemical warfare agent detector in the literature is developed, optimised and qualified on compounds that are not chemical warfare agents.[1,2] Dimethyl methylphosphonate stands in for the G-series nerve agents,[3] dipropylene glycol monomethyl ether for training detectors,[4] and methyl salicylate for vapour penetration through protective ensembles.[5,6] The substitution is unavoidable, because the agents are lethal at concentrations where handling them requires a licensed facility, and it is load-bearing: a sensor reported as selective for sarin has, in most cases, never been exposed to sarin.

What licenses the substitution is a claim of similarity. The compound chosen is said to be agent-like, and similarity is read from molecular descriptors: vapour pressure, dipole moment, frontier orbital energies, substructure and shape. The claim that matters for a detector is different. It is that the substitute produces the response the agent would produce in the same device. Those two claims are connected only if descriptor similarity transfers to response similarity, and that transfer has not been measured.

It has not been measured because measuring it requires the agents. Response data on authentic agents exist in few places, and the compounds used as simulants are rarely run

alongside them. A recent survey of the field found no standard for chemical warfare agent simulants at all, and recorded that they have generally been selected on the basis of their historical use.[7] Our institute is an OPCW designated laboratory, which identifies and assays authentic agents under an accredited procedure, so the comparison can be made on the agents themselves.

This work reports that measurement. A 29-dye colorimetric array was used to record the responses of sixteen authentic agents at four exposure concentrations in two optical channels, and those responses were compared against the molecular-descriptor distances that simulant selection relies on. The array separates compounds in every run and channel, so the measurement is sound. Descriptor distance nonetheless carries no detectable information about response distance. Dimethyl methylphosphonate, the compound the field treats as the G-agent surrogate, is anti-correlated with two of the four G agents, and its resemblance to sarin weakens as the concentration falls towards the range where point detectors operate. Where the designation record and a regulatory toxicity model might have supplied the missing evidence, both are empty for the same compounds.

This is not a new sensing method, and we propose no replacement compound. It is a measurement of whether the substitutes the field already relies on carry the property they are relied upon for.

Results and Discussion

The array separates the agents

A comparison between descriptor space and response space is worth making only if the response measurement resolves the compounds at all. That question is already answered in the reports these data come from, and the answer does not depend on anything we do here. On the first run a linear discriminant classifier built on a six-dye subset under dual illumination separated the four G agents, VX and dimethyl methylphosphonate at 100%

accuracy, against 87.5% under visible illumination alone.[8] On the second, a six-dye subset classified all seven agents at an accuracy of 0.964 with a macro F_1 of 0.964 under five-fold stratified cross-validation.[9] On the third, a four-dye subset separated the four Novichok agents at an accuracy of 1.000 and a macro F_1 of 1.000 under leave-one-out cross-validation, over a working range of 10 to 500 μM with a detection limit near 10 μM.[10]

Three qualifications belong with those figures, and together they move where the premise rests. Each is obtained on a dye subset chosen for that run, not on the 29-dye array we analyse. The first run reports no held-out evaluation, so its 100% is an apparent rather than a validated accuracy. And the run that carries the comparison this paper is about is that first run, because it is the only one in which a designated simulant was measured beside the agents; the two cross-validated figures come from the second and third runs, which contain no simulant. A validated accuracy on a run that holds no simulant does not establish the premise for a run that does. We also cannot reproduce either cross-validated figure from the deposited tables, which are averaged over replicates, so we quote them and do not verify them. The premise for the first run therefore rests on our own check below, which covers all six run-channel cells including that run's two.

What our own check adds is narrower and is reported because the comparison that follows is made on patterns rather than on class labels. Each exposure gives a 29-dye ΔE_{00} vector, which we centred dye-wise at each concentration and normalised to unit length, so that agreement is measured between patterns and not between magnitudes, and took agreement as the cosine. The question it answers is whether a compound's pattern is stable enough across dose to be treated as that compound's pattern at all.

It is stable enough (Figure 1a). One compound's patterns at two concentrations correlate at $+0.345$ to $+0.542$, different compounds anti-correlate at -0.103 to -0.330, and the ordering holds in all six run-channel cells, with p from 4.6×10^{-8} to 0.025.

The tables analysed here hold one value per compound, concentration, dye and channel, so the within-compound term is agreement of a compound with itself at a different dose

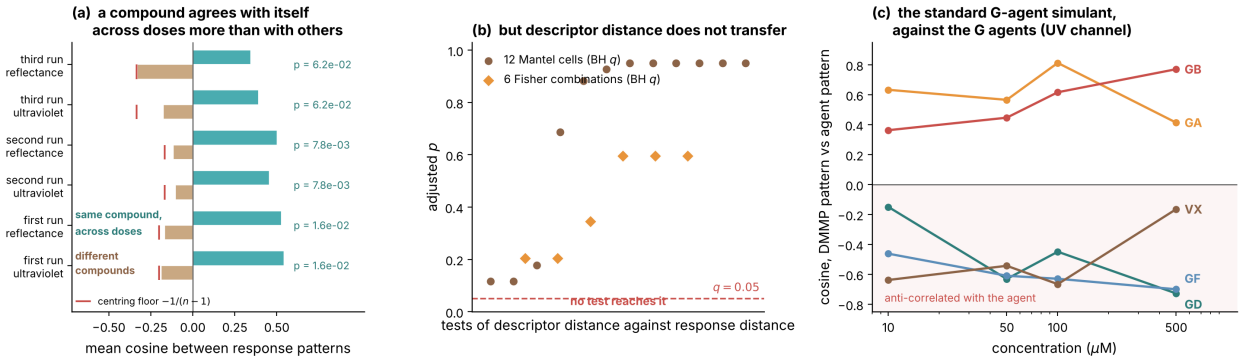


Figure 1: Response-space tests on authentic agents. (a) Mean cosine between response patterns, within one compound across concentrations and between different compounds, for three runs in two optical channels. (b) Adjusted p for every test of descriptor distance against response distance: twelve exact Mantel cells and six Fisher combinations of the independent runs. (c) Cosine between the response pattern of DMMP and those of the four G agents and VX, by concentration, in the ultraviolet channel.

and not agreement between repeated measurements. Repeats were reported in the source work for the second and third runs, for selected dye-analyte combinations in the second[9] and for all experiments in the third,[10] but the repeat-level values are not in these tables, so no repeatability estimate enters any analysis below.

Two features of the measurement belong here rather than in the Supporting Information. The first is that dye-wise centring forces the mean between-compound cosine towards an algebraic floor of $-1/(n-1)$, and the observed means clear that floor by between 0.0032 and 0.1576 across the six cells. The smallest margin is the reflectance channel of the third run, where four compounds put the floor at $-1/3$ and the observed mean is -0.3302. In that cell the between-compound structure is close to what centring alone would produce, and we do not read a separation from it. Most of the apparent separation in the other cells is a property of the centring too, and what our test on its own establishes is the weaker statement that a compound agrees with itself across concentrations more than it agrees with the other compounds in its cell on average. It does not establish the stronger pairwise statement: in 13 of the 34 compound-and-cell instances, some other compound's consensus pattern is closer to a compound than that compound's own agreement across concentrations. The classification

results reported for these runs in the source work, not this cosine, are what establish that the agents are separable, and we quote those rather than reproduce them, because the deposited tables are averaged over replicates and the cross-validation design cannot be rebuilt from them. The second is that six dye columns in the ultraviolet pivot of the second run were bit-identical across every row. Tracing them to the source images showed that those dyes do not fluoresce under ultraviolet excitation and the regions of interest had read plate background, so one channel had been recorded six times. Columns identical across all rows are counted once, which leaves 23 effective channels in that cell and 25 in the ultraviolet pivot of the third run. Supporting Information Section S19 sets out what the array does and does not establish in more detail.

Descriptor distance does not predict response distance

Simulant selection rests on an implicit transfer: that a compound close to an agent in descriptor space will be close to it in whatever the detector measures. We tested the transfer directly, as the agreement between two distance matrices over the same compounds, one built from descriptors and one from response patterns. Agreement was measured by Mantel correlation and referred to a null built by enumerating every relabelling of the compounds, so the test is exact rather than sampled.

The test is run within each run and never across runs. The three runs share no compound, so the dye-wise sensitivity of one plate cannot be calibrated against another, and pooling responses across them would require a commutability assumption we cannot support. That restriction sets the power. Six compounds in the first run and six in the second carry both a response pattern and a descriptor vector, giving fifteen pairs in each, and the third run drops out because the A-series compounds are not in the descriptor panel. Nitrogen mustard drops out of the second run for the same reason.

Across the twelve cells, three spaces by two channels by two runs, nothing survives correction (Figure 1b). Three cells reach a nominal p below 0.05, all of them in the second

run: the structural space in the ultraviolet channel at 0.0167, the merged space in the same channel at 0.0194, and the electronic space in the reflectance channel at 0.0444. Benjamini–Hochberg correction over the twelve leaves a smallest adjusted value of 0.1167. The pattern of the three is itself uninformative about transfer, because the two ultraviolet cells implicate the structural and merged spaces while the reflectance cell implicates the electronic space, and the same space does not reproduce across channels of the same plate.

There is a further asymmetry in where those three cells sit, and it matters more than their size. The second run contains seven authentic agents and no designated simulant. Whatever agreement those three cells carry is agreement between one agent and another, measured on one plate, and it is not evidence that a simulant reproduces an agent. The only run that contains a designated simulant beside the agents is the first, and the first run gives nothing in any space or channel, with $|\rho| \le 0.225$ and every p at or above 0.229. The one run that could speak to simulant fidelity is silent, and the run that speaks is not about simulants.

Combining runs does not change the answer. The first and second runs share neither compounds nor plates, so their exact Mantel p values are independent and can be combined by Fisher's method. We report all six combinations of three spaces and two channels rather than selecting one, which would reintroduce the analyst freedom the test is meant to remove. The smallest adjusted value over the six is 0.204, and none reaches 0.05. Section S12 of the Supporting Information gives the full permutation family, and Section S9 the six multiplicity schemes under which it was read.

What this does not license is the claim that no transfer exists. Fifteen pairs is a weak test, and the pooled rank correlations are correspondingly imprecise: the structural space in the ultraviolet channel gives $\rho = +0.31$ with a 95% interval from -0.45 to $+0.81$, which excludes neither zero nor a strong association. The defensible statement is that the transfer on which simulant selection depends has not been established, in the one body of authentic-agent response data where it could be tested, and that the experiment which would settle it is a run containing a compound already measured in an existing run. Such a bridge would

fix the plate correction and open the 136 cross-run pairs that are closed today.

The standard G-agent simulant against the G agents

The transfer test above treats all compounds alike. One compound deserves to be named, because the field treats it as the G-agent surrogate and because decontamination efficiency is quantified against it. The first run contains DMMP together with the four G agents and VX, so its response can be set against theirs directly.

In the ultraviolet channel DMMP correlates with tabun at $+0.764$ and with sarin at $+0.645$, and anti-correlates with soman at -0.492, cyclosarin at -0.721 and VX at -0.702. In the reflectance channel the same comparison gives -0.112, $+0.569$, -0.479, -0.495 and -0.273. The mean over the four G agents is negative in both channels, -0.189 and -0.135. The compound used in place of the G agents does not, on this measurement, resemble the G agents as a class.

A weaker claim survives in one channel. If DMMP is G-like at all it should sit closer to the G series than to VX, and that contrast is $+0.513$ with a 95% interval from $+0.131$ to $+0.722$ in the ultraviolet channel, which excludes zero. In the reflectance channel the same contrast is $+0.137$ with an interval from -0.144 to $+0.290$, which does not. One optical channel supports the ordering and the other does not resolve it.

Concentration changes the picture (Figure 1c). The DMMP-to-sarin cosine rises monotonically from $+0.363$ at 10 μM to $+0.771$ at 500 μM, and the DMMP-to-cyclosarin cosine falls monotonically from -0.462 to -0.700 over the same range. Monotonicity is perfect in three of the ten agent-channel series. It is also the strongest result the design can give: with four concentrations the two-sided exact permutation has a floor of $p = 0.083$, three series reach exactly that floor, and Benjamini–Hochberg correction over the ten leaves a smallest adjusted value of 0.278. We report the direction and not a significance claim.

Two limits sit on that direction. The concentrations are of analyte in a dye solution, not of vapour at an inlet, so we do not translate the range into the operating band of a point

detector. And the mean response magnitude rises with concentration at a Spearman correlation of exactly +1.000 in every run-channel cell, so a similarity that grows with concentration cannot be separated here from one that grows with signal-to-noise ratio. Distinguishing the two needs a repeat-level noise estimate, which the deposited tables do not carry. What the direction supports is the narrower statement that the resemblance between this simulant and this agent is not constant over the range at which the array was characterised.

Two further readings suggested themselves to us and did not survive their own controls. We report both, because the alternative is to present artefacts as results.

The first was a ranking. DMMP is the nearest of the five candidates to tabun in the ultraviolet channel and to sarin in the reflectance channel, and it is nearest to nothing else, and the two channels never agree on which agent it is nearest to. That reads as evidence that the choice of optical channel decides which agent the standard simulant stands in for. It is not. With five candidates the probability of rank one is 1/5 under random assignment, the expected number of first places over ten agent-channel cells is 2.0, and the observed number is two, with a one-sided binomial p of 0.62. The ranking carries no information.

The second was a channel decomposition. The cosine is a sum over dye channels, and the DMMP-to-tabun value of +0.764 turns out to rest on two of the 29 channels, with the largest single channel carrying 44% of the positive mass. Read alone, that makes the resemblance look fragile. Two controls show it is not a property of DMMP. Agent-to-agent pairs in the same run concentrate identically, with median two channels in the ultraviolet and one in the reflectance channel, and a null built by permuting the dye labels of the DMMP vector gives a mean of 2.00 and 2.26. Concentration of this kind is what the product of two centred unit vectors in 29 dimensions does, and it says nothing about this compound.

What remains after both controls is the set of measured values themselves, and one structural fact. The first run contains no candidate simulant other than DMMP, so the designation cannot be contrasted against an alternative in response space at all. The permutation null that the descriptor analysis of the next section relies on is, in response space,

empty.

Descriptor space cannot supply the missing validation

If response space cannot establish the transfer, the question becomes whether descriptor space can at least establish the premise it is used for, that the incumbents sit closer to the agents than other compounds do. This can be asked exactly. Designating three compounds out of the non-agents in a panel is a choice among a finite set, so the incumbent designation can be referred to every alternative designation rather than to a sampled null. The primary panel of thirty-five compounds gives 2024 designations and the reduced panel of twenty-eight gives 680.

On the primary panel the answer is flat (Figure 2). No agent of the eleven has a valid match, the smallest nominal p over the eighteen tests is 0.1917, and the Benjamini–Hochberg adjusted value is 0.6299.

What produces that zero is worth stating, because it is not chemical background. The criterion has two parts, a normalised distance of 2 or less and a distance smaller than that to every compound outside the designated set, and on this panel it is the second part that fails. Sarin, soman and cyclosarin all sit inside the threshold of DMMP, at 1.1805, 1.4583 and 1.6041. Each of them fails because diisopropyl fluorophosphate is nearer still, and tabun, which is outside the threshold at 2.5320, has diethyl cyanophosphonate nearer than DMMP. Those two compounds are not interferents. They are compounds the sensing literature itself proposes as simulants. If the comparison set is narrowed to the fourteen interferents, which is how the second part is implemented in the deposited distance table, the same three G agents pass and the count on this panel is three rather than zero. We report the wider reading as the primary one because it is the criterion stated in Section , and we report both counts because the difference between them is not a rounding matter. The designated simulants are not failing to stand out from chemical background on this panel. They are being displaced by alternatives the literature has already put forward.

Nothing in that panel supports the designation of the incumbents in particular, and the reason it does not is that better candidates exist in descriptor space.

The reduced panel, which omits five compounds the literature also calls simulants, reads differently. Three agents acquire a valid match, and the merged space returns a nominal one-sided p of 0.0250 with the structural space at 0.0279. Whether that survives depends on which correction is applied. Benjamini–Hochberg over the eighteen tests gives 0.1632, correcting within each space gives 0.0824, and the Westfall–Young single-step procedure, which uses the shared permutations and so handles the dependence between tests exactly, gives 0.0794. Only the scheme that fixes one statistic in advance returns a value below 0.05, at 0.0419. Significance here is a property of one correction scheme on one panel construction, and we report all four rather than the one that passes.

The surviving structural value is also an ordinary draw from the choices available to an analyst. Drawing three descriptors at random from the structural block 2000 times and repeating the whole procedure places the observed value at the 22.4rd percentile of that distribution. A value at the quarter mark of what random descriptor choices produce is not evidence about chemistry. The null for that percentile, the dimensionality controls and the effect of weighting the two descriptor blocks equally are in Supporting Information Sections S13, S14 and S15.

One quantity in this family does separate the incumbents from the alternatives, and it is worth stating precisely because it is easy to over-read. Across 500 random three-descriptor subspaces, the incumbent designation reaches $p < 0.05$ in 36.0% of them. Referred to the same quantity computed for all 680 designations, where the mean is 4.7% and the median 2.0%, the incumbent designation ranks fifth of 680 ($p = 0.0074$). The incumbents are therefore unusual in how often a lucky choice of three descriptors can be found for them. That is a statement about the flexibility of the comparison, not about proximity to the agents.

One analyst freedom we did not enumerate when we set the comparison up moved the

answer further than any we did. The electronic descriptors throughout are GFN2-xTB, a method parameterised for geometries and energies rather than for virtual orbital energies. Recomputing all three coordinates of the electronic space at B3LYP/def2-SVPD moves it from one valid match of eleven and a one-sided p of 0.275 to five matches and 0.0368. Four of those five are pairings the designation record supports: sarin, soman and tabun matched to DMMP, and cyclosarin matched to DPGME. We do not promote that value to a positive finding. It is a single uncorrected statistic in one space, reached by changing a method after seeing the first answer, and the two sets of calculations sit on different geometries, so the difference carries a geometry term we cannot separate from the method term. What it establishes is narrower and bears directly on this paper's argument: the space in which the incumbents did worst is the space whose answer the choice of electronic-structure method decides. The merged space does not move at all: thirty-eight coordinates, three valid matches and a one-sided p of 0.0250 under both methods, to the precision the exact enumeration of 680 designations can report. The result we carry forward is therefore not an artefact of the semiempirical method, while the three-coordinate space is a function of it. Supporting Information Section S8 gives the full comparison and the limits on it.

Panel composition decides the answer as well. Adding back the five compounds the sensing literature also treats as simulants moves the count from three valid matches to zero, because diisopropyl fluorophosphate is nearer to three G agents than any incumbent is, and 2-chloroethyl ethyl sulfide is nearer to sulfur mustard than dichloromethane is. The incumbents do not move. What moves is whether anything closer is allowed into the comparison.

So does the way the descriptors are encoded, and that freedom is the largest of the four. Four of the thirty-eight retained coordinates are binary indicators carried by exactly one compound of the twenty-eight: an arsenic count for lewisite, a phosphorus–cyanide count for tabun, a phosphorus–sulfur count for VX and a nitro count for chloropicrin. A binary column with a single one has sample variance exactly $1/n$, so standardising it puts the carrier and every non-carrier $\sqrt{n}$ apart and adds exactly n, here 28, to the squared distance between

that one compound and every other. The addition says that the compound is unique in the panel, not that it is far from DMMP, and for tabun it is 47.6% of the squared distance to DMMP. Dropping those four coordinates moves tabun from 2.0038 to 1.4509 and the count from three valid matches to four. Dropping every binary indicator carried by three compounds or fewer gives five, and dropping every coordinate with five or fewer distinct values gives six, with the merged-space one-sided p falling from 0.0250 to 0.0074. We kept the encoding that produces the smallest count, and we report the sweep because that is the only evidence that we did not select the encoding for its answer. Supporting Information Section S6 gives the column-by-column accounting.

Four analyst choices therefore move the count of valid matches: the panel, the order in which the correlation filter is handed its columns, the conformer embedding seed and the descriptor encoding. None of them moves the result of Section . Recomputing the twelve Mantel tests under the encodings above leaves the smallest Benjamini–Hochberg value at 0.1167, 0.4722 and 0.3889, none below 0.05, and removing the single-carrier indicators makes the agreement weaker rather than stronger. We draw the distinction explicitly because the two results are not equally load-bearing. The agreement test between descriptor distance and measured response survives every choice we enumerated. The count of valid matches survives none of them, and a quantity that four arbitrary choices can move between three and six is not a measurement of anything. That is this paper's argument applied to its own instrument, and it is the reason we do not report a match count as a finding.

Neither the designation record nor a toxicity model fills the gap

Two places remain where the missing evidence might be found, and both are empty for the same compounds.

The first is the standard-setting record. We traced each incumbent to the document that designates it. DMMP is identified in a NIOSH programme for respirator materials resistance testing, where the measurement is filter breakthrough and the same document states that

the simulant information is not to be used as a recommendation of any simulant.[3] DPGME appears in a US Army patent for a training composition, where the property required of it is detectability by ion mobility spectrometry.[4] For dichloromethane we found no designating document in the public record. Each of the two traceable designations is a validation against a measurement, and neither of those measurements is detection. A simulant is validated for a purpose, and that validation does not travel with the compound to a different purpose.

The second is regulatory toxicity (Figure 3). One of the three incumbents carries a regulator-derived acute inhalation value. DMMP, the compound against which decontamination efficiency is quantified, carries none of any kind. A model cannot fill that gap. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ in cross-validation, with an out-of-fold bias of +0.001 log units, so nothing in its internal statistics warns a user off. Applied to twenty military agents held out of training it is wrong in one direction without exception, by 0.40 to 3.42 log units. Its applicability-domain check certifies fourteen of the twenty as inside the domain, and the worst of those fourteen is wrong by 2.68 log units, a factor of 482. The mean error over the twenty is +1.68 log units with Cohen's $d_z = +1.96$. The measured axis and its gaps are tabulated in Supporting Information Section S21, and the test of whether fidelity and acute toxicity are associated at all, which finds no monotonic relation across the compounds carrying both axes, is in Section S22.

Conclusions

Simulant validity is not a property of a compound. It is a claim about a measurement, and the practice of designating chemical warfare agent simulants records neither the measurement a compound was validated against nor the comparison in which its validity was tested.

Three counts fix where the evidence stands. In the three runs reanalysed here, one designated simulant was measured beside the agents, in one run of one array, and we found no second co-measurement in the literature we searched; we state that as the limit of our

search rather than as a claim about every measurement ever made. Twelve exact Mantel tests and six combinations of them leave no agreement between descriptor distance and response distance below $q = 0.05$, against a design that resolves a rank correlation of 0.67 at 80% power. On the primary panel no agent of the eleven has a valid match to an incumbent simulant, at a Benjamini–Hochberg adjusted value of 0.6299, and the compounds that displace the incumbents there are themselves literature-proposed simulants rather than interferents.

The documentary record accounts for the first of those counts rather than contradicting it. DMMP was designated for respirator filter breakthrough testing and DPGME for an ion-mobility training composition. Neither measurement is detection, and a validation does not travel with a compound to a purpose it was never tested for. The toxicity axis is empty in the same place: DMMP, the compound against which decontamination efficiency is quantified, carries no regulator-derived acute inhalation value, and a random forest trained on 230 AEGL chemicals is wrong in one direction without exception on twenty held-out military agents, so a prediction cannot stand in for the missing measurement.

Absence of evidence is not evidence of unsuitability, and we conclude nothing about whether the incumbents are poor choices. We propose no replacement compound either, because selecting one on descriptor similarity would repeat the error this paper is about.

Two readings this paper does not support should be ruled out, because the designating documents are cited here for what they measured. DMMP is the benchmark compound against which decontamination efficiency is quantified, and nothing in this work bears on whether it is suitable for that. Filter breakthrough and surface decontamination are mass-transport and reactivity measurements, our comparison is between a colorimetric response and a descriptor vector, and a compound can be a sound surrogate for one and an unsound surrogate for the other. The same holds for DPGME and ion mobility spectrometry: the patent that designates it requires detectability by that technique, we did not measure ion mobility, and our result says nothing about whether the compound serves that purpose.

The claim at issue is narrower than either. It is that a validation performed against one measurement is being carried, without a second validation, to colorimetric and spectroscopic detection of the agents themselves.

What can be specified is the measurement that would settle the question. It is a bridging run: candidate simulants and the agents they stand in for exposed on the same plate, in the same optical channels, at the same concentrations, within one session. Such a run yields the one quantity no amount of descriptor work can supply, a response-space distance between a simulant and its agent measured under identical conditions, and it needs no new chemistry and no new instrument. It needs only that the simulant be placed beside the agent rather than in place of it. Until that run exists, a statement that a compound is a validated detection simulant reports a convention, and a convention should be declared as one.

Experimental Section

Array measurements

The response data are the colorimetric tables of three published runs of one array, reanalysed here and not remeasured. The first covers tabun, sarin, soman, cyclosarin, VX and dimethyl methylphosphonate;[8] the second hydrogen cyanide, cyanogen chloride, phosgene, chloropicrin, sulfur mustard, nitrogen mustard and lewisite;[9] the third the Novichok agents A-230, A-232, A-234 and A-242.[10] All agents were produced and handled at the CBRN Defense Research Institute, Seoul, an Organisation for the Prohibition of Chemical Weapons designated laboratory, under Chemical Weapons Convention procedures. For the second run, purity was confirmed by GC–MS and exceeded 95% for every compound except lewisite, at 67%.[9] The first and third runs report no purity values, and we have not been able to supply them; the reader should treat the purity of those compounds as unstated rather than as high. Dropping lewisite changes no count reported here, on either panel and in either optical channel of the second run (Supporting Information Section S7).

The array is the same 29 commercial chromogenic and fluorogenic dyes in the same order in all three runs, commercially sourced and used as received. Stock solutions at 100 mM were prepared in ethanol, methanol, dimethyl sulfoxide, water, tetrahydrofuran or toluene and sonicated to dissolution, then diluted into acetonitrile to a working concentration of 100 μM, which holds the residual stock solvent below 0.1% (v/v) and places every dye in one solvent environment. Each analyte was delivered in acetonitrile by serial dilution to final concentrations of 0, 10, 50, 100 and 500 μM in a 2.0 mL dye solution. Samples were mixed without incubation and imaged at once, under visible light and under ultraviolet excitation at 365 nm, with a smartphone camera fixed in an enclosed imaging box at 22 $\pm$ 3 °C and 35 $\pm$ 5% relative humidity. Regions of interest were central 20 $\times$ 10 pixel zones; RGB values were read at five positions within each region, averaged, converted to CIE $L^*a^*b^*$, and expressed as the CIEDE2000 colour difference ΔE_{00} against the 0 μM image of the same dye in the same channel. The self-referencing removes illumination and device offsets common to an image pair.

The three runs are different compounds measured on different plates and are not replicates of one another, so no ΔE_{00} value is compared across runs; every analysis below is run within a run and channel, and results are combined only at the level of correlation coefficients. The two channels are physically different measurements rather than repeats and are treated as independent modalities. The tables hold one value per compound, concentration, dye and channel.

Response patterns

Each exposure gives a 29-dye ΔE_{00} vector. Vectors were centred dye-wise within each concentration and normalised to unit length, so that agreement is measured between patterns and not between magnitudes, and agreement was taken as the cosine. Dye columns that are bit-identical across every row carry no information and were counted once, which leaves 23 effective channels in the ultraviolet pivot of the second run and 25 in that of the third.

Descriptors

Structural and physicochemical descriptors, 51 after excluding run metadata, were computed with RDKit from canonical SMILES. Ten are read from a three-dimensional structure and are taken as Boltzmann-weighted means over an ETKDGv3 conformer ensemble at 298 K with MMFF94 energies, or UFF for lewisite 1, the one compound without MMFF parameters, so that no result depends on an undeclared choice of conformer; Supporting Information Section S4 reports what the two treatments do to the results. Electronic descriptors were computed with GFN2-xTB in the gas phase, Boltzmann averaged over up to five conformers at 298.15 K. The electronic space carries E_{HOMO}, E_{LUMO} and the dipole moment. The Parr electrophilicity index $\omega = \mu^2/2\eta$ is reported but admitted to no distance, because it is an exact function of the two orbital energies that accompany it and a filter built on linear correlation does not detect that. It and the condensed Fukui index at phosphorus, which exists only for the six phosphorus-bearing compounds, are reported in Supporting Information Section S18 and admitted to no distance. Dipole moments were recomputed at B3LYP/def2-SVPD, together with the two frontier orbital energies. The merged space, which carries the result reported above, is unchanged by that substitution. The three-coordinate electronic space is not, and we report what it does in Section rather than leaving the check described as a confirmation (Supporting Information Section S8). Identifiers and formulae were resolved against PubChem, and no property value was taken from a prediction service.

Distances and tests

Within each space, descriptors of zero variance were removed, then one of each pair correlated at $|r| > 0.95$, retaining a protected set. Remaining descriptors were standardised and distances taken as Euclidean, normalised by the median nearest-neighbour separation of the same space, which makes a normalised distance comparable within a panel and not between panels.

A paired comparison of the distance to the nearest incumbent simulant against the dis-

tance to the nearest interferent is confounded by group size, since the minimum of many draws is smaller than the minimum of few by construction. We therefore enumerated every way of designating three of the non-agents as simulants and report the rank of the actual designation as an exact p-value: $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The hypothesis is directional, so the reported p is one-sided. We give the two-sided values here rather than only the convention, because on the twenty-eight-compound panel they decide the answer: the structural space moves from 0.0279 one-sided to 0.0559 at twice the smaller tail and 0.0441 scored as distance from the null mean, and the merged space from 0.0250 to 0.0500 and 0.0279. Under the first of the two-sided conventions neither space reaches 0.05, and the one correction scheme that returns a value below 0.05 does so only on the one-sided reading. The Mantel tests of Section are two-sided throughout, since the direction of a matrix correlation is not fixed in advance by the hypothesis. A valid match requires a normalised distance of 2 or less together with a distance smaller than that to every compound outside the designated set. Sensitivity to descriptor choice was assessed by resampling the merged-space descriptors with replacement 1000 times under a fixed seed, restandardising and recomputing within each replicate; the intervals so obtained are percentile ranges over descriptor choice and not confidence intervals in the sampling sense.

Agreement between descriptor distance and response distance was tested by Mantel correlation with exact enumeration of relabellings in each run-channel cell, with the independent runs combined by Fisher's method and the family of tests corrected by the Benjamini–Hochberg procedure. Six distance statistics and six multiplicity schemes were carried through in parallel so that the conclusion can be read against the analyst choices rather than from one of them.

Software

NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, Matplotlib 3.11.2, RDKit 2026.03.6 and PySCF 2.14.0. All analyses are deterministic given the seed, and the permutation tests are exhaustive rather than sampled. Analysis code, descriptor tables, the parsed regulatory values with their sources and every figure script are deposited in a public repository,[11] from which every number reported here can be regenerated.

Acknowledgement

The colorimetric response data reanalysed here were measured in three earlier studies from this laboratory,[8–10] and we thank the staff of the CBRN Defense Research Institute who performed those measurements under Chemical Weapons Convention procedures. K.K. carried out the conceptualisation, software, formal analysis and visualisation and wrote the original draft. J.K. and J.Y. contributed to the analysis and validation, J.Y. to the original draft, M.K. to the validation and to reviewing and editing, D.-H.L. to validation, resources and supervision, and Y.J.J. to validation, supervision and project administration. This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. The authors declare no competing financial interest.

Supporting Information Available

The Supporting Information is available free of charge at `https://pubs.acs.org/doi/XX.XXXX/acssensors.XXXXXXX`.

- Acute inhalation toxicity of the panel compounds and the gaps in it; predictive interpolation of that axis and why a model cannot fill the gaps; the decoupling test between fidelity and acute toxicity; conformer dependence of the shape descriptors and the reproduction gate; the density-functional check on the electrophilicity axis and on the

dipole; the full permutation family under all six multiplicity schemes; dimensionality, subspace and descriptor-block weighting controls, each with its null; the measured colorimetric array response and what it does and does not establish; phosphorus-local reactivity for the six phosphorus-bearing compounds; and the structural filter (PDF)

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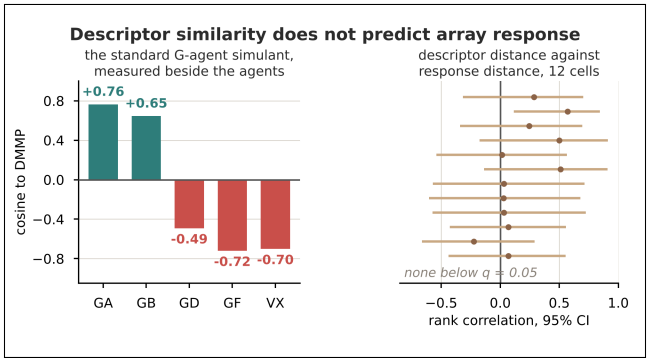
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TOC Graphic


Descriptor similarity does not predict array response
the standard G-agent simulant, measured beside the agents
cosine to DMMP
0.8
0.4
0.0
−0.4
−0.8
+0.76
+0.65
-0.49
-0.72
-0.70
GA
GB
GD
GF
VX
descriptor distance against response distance, 12 cells
none below q = 0.05
−0.5
0.0
0.5
1.0
rank correlation, 95% CI

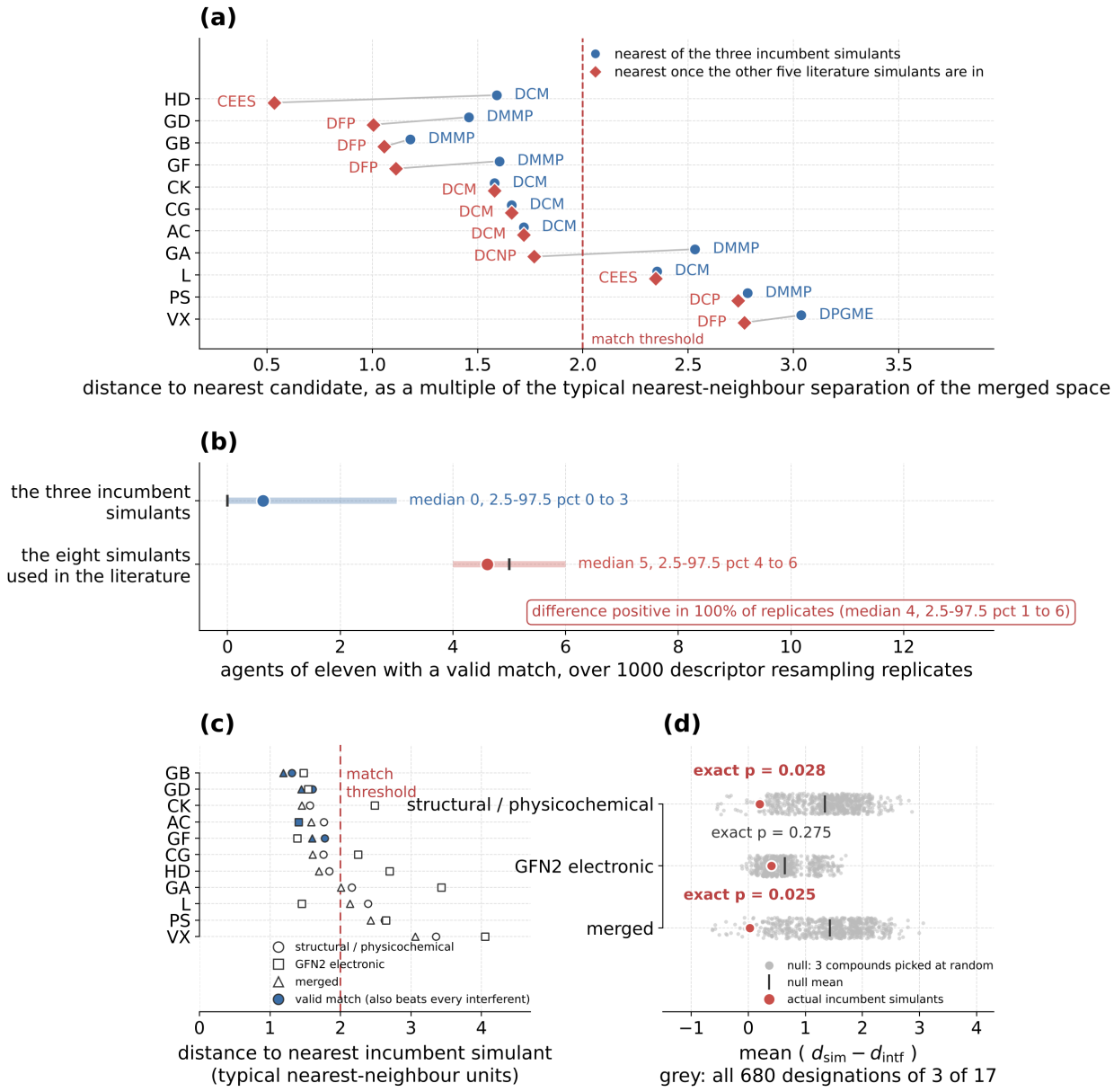


Figure 2: Descriptor-space tests. (a) Distance from each agent to its nearest candidate, for the three incumbents and for the eight compounds the sensing literature also calls simulants. (b) Valid matches over 1000 descriptor resampling replicates. (c) Distance to the nearest incumbent by space. (d) Observed mean difference against the exact null over all 680 designations of three compounds from seventeen.

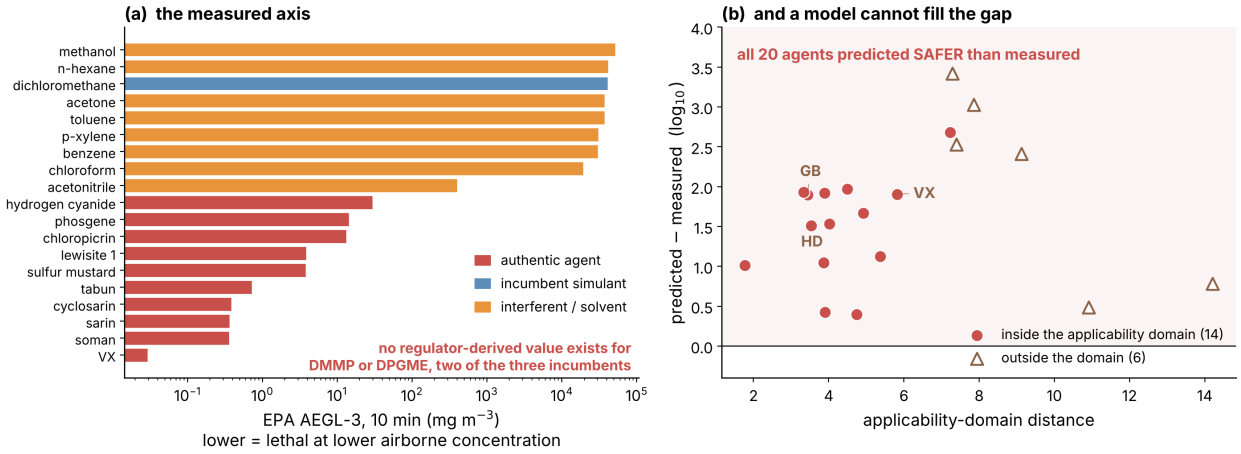


Figure 3: The toxicity axis. (a) Measured EPA AEGL-3 10-minute values for the panel compounds that carry one. (b) Prediction error of a random forest trained on 230 AEGL chemicals, for twenty military agents held out of training, against distance to the training set.